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DEEMED TO BE
UNIVERSITY
NAVI MUMBAI

Semester: IV

B. Tech. Computer Science and Engineering (Artificial Intelligence and Machine Learning)

38222401

CAC401

Artificial Intelligence

Date: 03-06-2025

Time: 10.00 AM TO 12.00 NOON

Marks: 60

Q.1 Solve any Four

20 Marks

CO

BT

- a) What are PEAS descriptors? Give PEAS description for Part - Picking Robot.
- b) Consider the undirected graph with starting position at node 0. List the nodes in the order of visit using Breadth First Search (BFS)

5

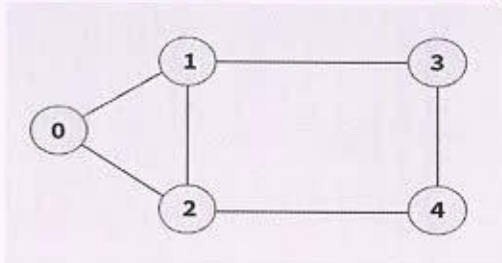
CO1

BT3

5

CO2

BT3



- c) Using Game theory "Tic Tac Toe" explain Min -Max Search?
- d) Explain Forward Chaining algorithm with the help of example.
- e) Compare and contrast classical logic and fuzzy logic.
- f) Write a short note on Hierarchical Planning.

5

CO3

BT3

5

CO4

BT2

5

CO5

BT2

5

CO6

BT2

Q.2 Solve any Four

20 Marks

CO

BT

- a) Define artificial intelligence. What are different AI applications. Enlist five AI applications.
- b) Analyze depth limited search algorithm and give time and space complexity.
- c) Write a short note on the genetic algorithm.
- d) What is propositional logic? Write syntax and semantics and example sentences for propositional logic.
- e) A fuzzy set is given by $B = \{(1,0.1), (2,0.2), (3,0.3), (4,0.9), (0,0.0)\}$. What is the crisp set that can be concluded from it?

5

CO1

BT2

5

CO2

BT4

5

CO3

BT2

5

CO4

BT3

5

CO5

BT3

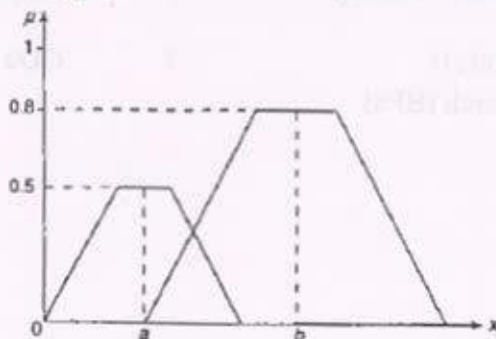


- f) Discuss partial order planning for wearing shoe example. 5 CO6 BT2

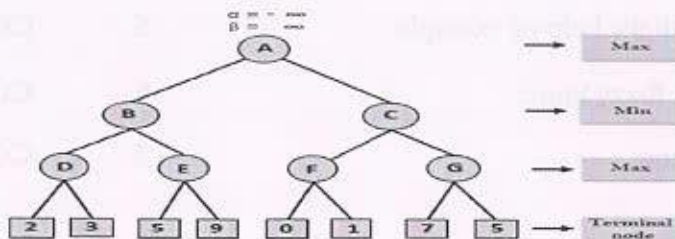
Q.3 Solve any Two

20 Marks CO BT

- a) Give the architecture of Fuzzy Logic System. What is Defuzzification method? Enlist any two Defuzzification methods. With the help of weighted average method calculate the defuzzified output for the fig. given below, where fuzzy sets 1 and fuzzy set 2 are considered as 50 and 70 resp.



- b) Evaluate optimal value using Alfa Beta alpha-beta pruning of following example. 10 CO3 BT5



- c) Using a predicate logic convert the following sentences to predicates and prove that the statement "Ram did not jump" is false.
- Ram went to temple.
 - The way to temple is, walk till post box and take left or right road
 - The left road has a ditch.
 - Way to cross the ditch is to jump.
 - Along is across the right road.
 - One needs to jump across the long to go ahead.